

# ELITE IGCSE ACADEMY

*Student Past Paper Solutions*

## May 2024 P1H Worked Solutions

May 2024 | P1H

31 questions   ■   question image followed by worked solution

PREPARED BY

**Dr. Eslam Ahmed**

*Assistant Lecturer, Cairo University Faculty of Engineering*

WhatsApp: +20 112 000 9622   |   eliteigcse.com

**PAST PAPER QUESTION****Q#: 1****Marks: 3**TOPIC: **Sequences**

May 2024 P1H - May 2024 | P1H

- 1 Here are the first four terms of an arithmetic sequence.

1          4          7          10

- (a) Find an expression, in terms of  $n$ , for the  $n$ th term of this sequence.

.....  
(2)

The  $n$ th term of a different arithmetic sequence is  $5n + 17$

- (b) Find the 12th term of this sequence.

.....  
(1)

(Total for Question 1 is 3 marks)

**PAST PAPER SOLUTION****Q#: 1****Marks: 3**TOPIC: **Sequences**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Sequences. The tag is correct.**Solution**

The sequence is

$$1, 4, 7, 10$$

The common difference is 3.

$$n\text{th term} = 1 + 3(n - 1) = 3n - 2$$

For the other sequence,

$$5n + 17$$

The 12th term is

$$5(12) + 17 = 77$$

**FINAL ANSWER** $3n - 2$ , and 77.

**PAST PAPER QUESTION****Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

May 2024 P1H - May 2024 | P1H

- 2 450 students were asked how they travelled to school on Monday.  
Each student walked or travelled by bus or travelled by car or travelled by bicycle.  
Each student used just one method of travel.

One of these students is chosen at random.

The table shows information about the probability of each method of travel.

| Method of travel | walk | bus | car  | bicycle |
|------------------|------|-----|------|---------|
| Probability      | 0.20 | $x$ | $2x$ | 0.26    |

Work out how many of the 450 students travelled by car.

(Total for Question 2 is 4 marks)

**PAST PAPER SOLUTION****Q#: 2****Marks: 4**TOPIC: **Probability Toolkit**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Probability Toolkit. The tag is correct.**Solution**Let the probability of travelling by bus be  $x$ .Then the probability of travelling by car is  $2x$ .

$$0.20 + x + 2x + 0.26 = 1$$

$$3x = 0.54$$

$$x = 0.18$$

So

$$P(\text{car}) = 2x = 0.36$$

Number travelling by car:

$$450 \times 0.36 = 162$$

**FINAL ANSWER**

162.

**PAST PAPER QUESTION****Q#: 3****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

May 2024 P1H - May 2024 | P1H

- 3 Find the highest common factor (HCF) of 72 and 108  
Show your working clearly.

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(Total for Question 3 is 2 marks)

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**PAST PAPER SOLUTION****Q#: 3****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Prime factors, HCF and LCM. The tag is correct.**Method**

$$72 = 2^3 \times 3^2$$

$$108 = 2^2 \times 3^3$$

For the HCF, take the smaller power of each common prime:

$$\text{HCF} = 2^2 \times 3^2$$

$$2^2 \times 3^2 = 4 \times 9 = 36$$

**FINAL ANSWER**

36

**PAST PAPER QUESTION****Q#: 4****Marks: 3**TOPIC: **Percentages**

May 2024 P1H - May 2024 | P1H

- 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

(Total for Question 4 is 3 marks)

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**PAST PAPER SOLUTION****Q#: 4****Marks: 3**TOPIC: **Percentages**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

April is 15% more than March, so April is 115% of March.

$$1.15 \times \text{March} = 943$$

$$\text{March} = \frac{943}{1.15} = 820$$

**FINAL ANSWER**

820 kilometres

**PAST PAPER QUESTION****Q#: 5****Marks: 4**TOPIC: **Angles in Polygons & Parallel Lines**

May 2024 P1H - May 2024 | P1H

- 5 In the diagram,  $ABCDE$  is a regular pentagon.

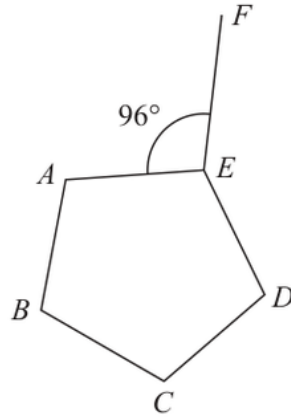


Diagram **NOT**  
accurately drawn

Angle  $AEF = 96^\circ$

Work out the size of the obtuse angle  $FED$

Show your working clearly.

o

(Total for Question 5 is 4 marks)

**PAST PAPER SOLUTION****Q#: 5****Marks: 4****TOPIC:** Angles in Polygons & Parallel Lines

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Angles in Polygons & Parallel Lines. The tag is correct.**Solution**

The interior angle of a regular pentagon is

$$\frac{(5 - 2) \times 180}{5} = 108^\circ$$

At  $E$ , angles around a point add to  $360^\circ$ .

$$\angle FED = 360^\circ - 96^\circ - 108^\circ = 156^\circ$$

**FINAL ANSWER**

156°.

**PAST PAPER QUESTION****Q#: 6****Marks: 5****TOPIC: Solving Linear Equations**

May 2024 P1H - May 2024 | P1H

- 6 (a) Expand and simplify  $(m + 5)(m - 8)$

.....  
(2)

- (b) Solve  $3n - 4 = \frac{5n + 6}{3}$

Show clear algebraic working.

 $n =$  .....  
(3)

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(Total for Question 6 is 5 marks)

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**PAST PAPER SOLUTION****Q#: 6****Marks: 5****TOPIC: Solving Linear Equations**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to solving linear equations. The original tag was expanding brackets, but the linear equation is the larger part.

**Solution**

(a)

$$(m + 5)(m - 8) = m^2 - 3m - 40$$

(b)

$$3n - 4 = \frac{5n + 6}{3}$$

$$9n - 12 = 5n + 6$$

$$4n = 18$$

**FINAL ANSWER**(a)  $m^2 - 3m - 40$ , (b)  $n = \frac{9}{2}$

**PAST PAPER QUESTION****Q#: 7****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

May 2024 P1H - May 2024 | P1H

- 7  $\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$   
 $A = \{\text{even numbers}\}$   
 $B = \{23, 29, 31\}$   
 $C = \{\text{multiples of 3}\}$

(a) List the members of the set

(i)  $B \cup C$ 

(1)

(ii)  $A' \cap C$ 

(1)

(b) Is it true that  $B \cap C = \emptyset$ ?

Tick (✓) one of the boxes below.

Yes

No

☐☐

Give a reason for your answer.

(1)

The set  $D$  has 4 members and is such that  $D \cap (A \cup C) = \emptyset$ (c) List the members of set  $D$ 

(2)

**(Total for Question 7 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 7****Marks: 5****TOPIC: Set Notation & Venn Diagrams**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Set notation and Venn diagrams. The tag is correct.**Method**

$$\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$$

$$A = \{24, 26, 28, 30, 32, 34\}$$

$$B = \{23, 29, 31\}$$

$$C = \{24, 27, 30, 33\}$$

So

$$B \cup C = \{23, 24, 27, 29, 30, 31, 33\}$$

Also  $A'$  is the set of odd numbers in  $\mathcal{E}$ , so

$$A' \cap C = \{27, 33\}$$

The statement  $B \cap C = \emptyset$  is true because none of 23, 29, 31 is a multiple of 3.For  $D$ , we need  $D \cap (A \cup C) = \emptyset$ . So  $D$  must contain the four numbers outside  $A \cup C$ :

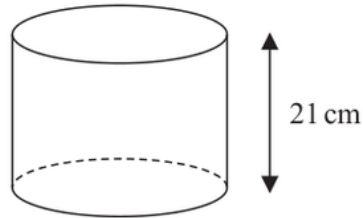
$$D = \{23, 25, 29, 31\}$$

**FINAL ANSWER**(a)(i)  $\{23, 24, 27, 29, 30, 31, 33\}$ , (a)(ii)  $\{27, 33\}$ , (b) Yes, (c)  $\{23, 25, 29, 31\}$

**PAST PAPER QUESTION****Q#: 8****Marks: 3****TOPIC: Standard & Compound Units**

May 2024 P1H - May 2024 | P1H

- 8 A cylinder is placed on a table.

Diagram **NOT**  
accurately drawn

The volume of the cylinder is  $1575 \text{ cm}^3$

The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

..... newtons/cm<sup>2</sup>

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**(Total for Question 8 is 3 marks)**



**PAST PAPER SOLUTION****Q#: 8****Marks: 3****TOPIC: Standard & Compound Units**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Standard & Compound Units. The tag is correct.**Solution**

The volume of a cylinder is

$$\text{volume} = \text{area of base} \times \text{height}$$

So the area in contact with the table is

$$1575 \div 21 = 75 \text{ cm}^2$$

Use

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$\text{pressure} = \frac{84}{75} = 1.12$$

**FINAL ANSWER**1.12 newtons/cm<sup>2</sup>.

**PAST PAPER QUESTION****Q#: 9** **Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2024 P1H - May 2024 | P1H

- 9 The table gives the amount of rice produced by each of two countries in 2020

| Country   | Amount of rice (tonnes) |
|-----------|-------------------------|
| Indonesia | $3.5 \times 10^7$       |
| Argentina | $8.2 \times 10^5$       |

- (a) Write  $3.5 \times 10^7$  as an ordinary number.

.....  
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020  
Give your answer in standard form.

..... tonnes  
(2)

**(Total for Question 9 is 3 marks)**

**PAST PAPER SOLUTION****Q#: 9****Marks: 3****TOPIC: Powers, Roots & Standard Form**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Powers, roots and standard form. The tag is correct.**Method**

$$3.5 \times 10^7 = 35000000$$

Argentina produced

$$8.2 \times 10^5 = 820000$$

tonnes of rice.

Japan produced 6780000 more tonnes than Argentina, so

$$820000 + 6780000 = 7600000$$

In standard form,

$$7600000 = 7.6 \times 10^6$$

**FINAL ANSWER**(a) 35000000, (b)  $7.6 \times 10^6$  tonnes

**PAST PAPER QUESTION****Q#: 10****Marks: 4**TOPIC: **Algebraic Roots & Indices**

May 2024 P1H - May 2024 | P1H

**10** (a) Simplify  $(2p)^0$  where  $p > 0$ .....  
(1)

$$y^9 \times y^{-3} = y^n$$

(b) Find the value of  $n$  $n =$  .....  
(1)(c) Simplify fully  $(5a^4c^2)^3$ .....  
(2)

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**(Total for Question 10 is 4 marks)**

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**PAST PAPER SOLUTION****Q#: 10****Marks: 4**TOPIC: **Algebraic Roots & Indices**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Algebraic roots and indices. The tag is correct.**Solution**

(a)  $(2p)^0 = 1$ .

(b)  $y^9 \times y^{-3} = y^{9-3} = y^6$ , so  $n = 6$ .

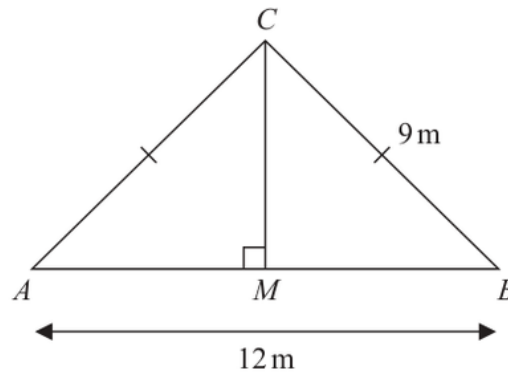
(c)  $(5a^4c^2)^3 = 5^3a^{12}c^6 = 125a^{12}c^6$ .

**FINAL ANSWER**

(a) 1, (b) 6, (c)  $125a^{12}c^6$

**PAST PAPER QUESTION****Q#: 11****Marks: 4****TOPIC:** Right-Angled Triangles - Pythagoras & Trigonometry

May 2024 P1H - May 2024 | P1H

**11** The diagram shows a roof support.Diagram **NOT**  
accurately drawnThe roof support is made from four lengths of wood,  $AB$ ,  $AC$ ,  $BC$  and  $MC$ 

$$AC = BC = 9 \text{ m} \quad AB = 12 \text{ m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.

Show your working clearly.

..... euros

**(Total for Question 11 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to Right-Angled Triangles - Pythagoras and Trigonometry. The missing support length is found using Pythagoras.

**Method**

Since  $AC = BC$ , the perpendicular support from  $C$  meets  $AB$  at its midpoint.

$$AM = \frac{12}{2} = 6$$

In right-angled triangle  $AMC$ ,

$$AC^2 = AM^2 + MC^2$$

$$9^2 = 6^2 + MC^2$$

$$MC^2 = 81 - 36 = 45$$

$$MC = \sqrt{45} = 6.708\dots$$

A whole number of metres must be bought, so buy 7 m for  $MC$ .

Total length needed:

$$12 + 9 + 9 + 7 = 37$$

Cost:

$$37 \times 21.50 = 795.50$$

**FINAL ANSWER**

795.50 euros.

**PAST PAPER QUESTION****Q#: 12****Marks: 5**TOPIC: **Algebraic Fractions**

May 2024 P1H - May 2024 | P1H

**12** (a) Factorise fully  $6y^2 - 5y - 4$ .....  
(2)(b) Express  $\frac{2x+1}{4x} + \frac{7-5x}{3x}$  as a single fraction in its simplest form......  
(3)

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**(Total for Question 12 is 5 marks)**

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**PAST PAPER SOLUTION****Q#: 12****Marks: 5****TOPIC: Algebraic Fractions**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Corrected to algebraic fractions. The fraction simplification is the larger part.**Solution**

(a)

$$6y^2 - 5y - 4 = (3y - 4)(2y + 1)$$

(b)

$$\frac{2x + 1}{4x} + \frac{7 - 5x}{3x}$$

Use denominator  $12x$ :

$$\begin{aligned} & \frac{3(2x + 1) + 4(7 - 5x)}{12x} \\ &= \frac{6x + 3 + 28 - 20x}{12x} = \frac{31 - 14x}{12x} \end{aligned}$$

**FINAL ANSWER**(a)  $(3y - 4)(2y + 1)$ , (b)  $\frac{31 - 14x}{12x}$



**PAST PAPER SOLUTION****Q#: 13****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Probability Diagrams - Venn & Tree Diagrams. The tag is correct.**Solution**

Bag A has 3 white and 7 black counters.

Bag B has 5 white and 4 black counters.

The same colour means both white or both black.

$$\begin{aligned}P(\text{same}) &= \frac{3}{10} \times \frac{5}{9} + \frac{7}{10} \times \frac{4}{9} \\&= \frac{1}{6} + \frac{14}{45} \\&= \frac{43}{90}\end{aligned}$$

**FINAL ANSWER**

$$\frac{43}{90}.$$

**PAST PAPER QUESTION****Q#: 14****Marks: 5**TOPIC: **Percentages**

May 2024 P1H - May 2024 | P1H

**14** The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are  $\frac{3}{2}$  times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

**(Total for Question 14 is 5 marks)**

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**PAST PAPER SOLUTION****Q#: 14****Marks: 5**TOPIC: **Percentages**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**Let Abel's savings be  $A$ .

Bahira has 45% more than Abel, so

$$B = 1.45A$$

Their combined savings are

$$A + 1.45A = 15435$$

$$2.45A = 15435$$

$$A = 6300$$

So

$$B = 1.45 \times 6300 = 9135$$

Bahira's savings are  $\frac{3}{2}$  times Chanda's savings, so

$$9135 = \frac{3}{2}C$$

$$C = 9135 \times \frac{2}{3} = 6090$$

**FINAL ANSWER**

6090 dinars

**PAST PAPER QUESTION****Q#: 15****Marks: 4****TOPIC: Functions**

May 2024 P1H - May 2024 | P1H

**15** The function  $f$  is defined as

$$f: x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of  $x$  that cannot be included in any domain of the function  $f$ .....  
(1)(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$  $f^{-1}(x) = \dots$   
(3)

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**(Total for Question 15 is 4 marks)**

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Dr. Eslam A.

**PAST PAPER SOLUTION****Q#: 15****Marks: 4****TOPIC: Functions**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Functions. The tag is correct.**Solution**

$$f(x) = \frac{3x + 1}{x - 2}$$

(a) The denominator cannot be zero, so

$$x - 2 = 0$$

$$x = 2$$

(b) Let

$$y = \frac{3x + 1}{x - 2}$$

$$y(x - 2) = 3x + 1$$

$$yx - 2y = 3x + 1$$

$$x(y - 3) = 2y + 1$$

$$x = \frac{2y + 1}{y - 3}$$

So

$$f^{-1}(x) = \frac{2x + 1}{x - 3}$$

**FINAL ANSWER**Excluded value 2, and  $f^{-1}(x) = \frac{2x + 1}{x - 3}$ .

**PAST PAPER QUESTION****Q#: 16****Marks: 4****TOPIC: Combined & Conditional Probability**

May 2024 P1H - May 2024 | P1H

**16** There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

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**(Total for Question 16 is 4 marks)**

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**PAST PAPER SOLUTION****Q#: 16****Marks: 4****TOPIC: Combined & Conditional Probability**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Combined & Conditional Probability. The tag is correct.**Solution**

At least one sweet of each colour is the complement of all red or all yellow.

$$P(\text{all red}) = \frac{15}{20} \times \frac{14}{19} \times \frac{13}{18}$$

$$P(\text{all yellow}) = \frac{5}{20} \times \frac{4}{19} \times \frac{3}{18}$$

$$P(\text{at least one of each}) = 1 - P(\text{all red}) - P(\text{all yellow})$$

$$= 1 - \frac{2730}{6840} - \frac{60}{6840}$$

$$= \frac{45}{76}$$

**FINAL ANSWER**

$$\frac{45}{76}$$

**PAST PAPER QUESTION****Q#: 17****Marks: 3**TOPIC: **Algebraic Proof**

May 2024 P1H - May 2024 | P1H

- 17 Show that  $\frac{1 + \sqrt{5}}{3 - \sqrt{5}}$  can be written in the form  $a + \sqrt{b}$  where  $a$  and  $b$  are integers.

Show each stage of your working clearly.

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(Total for Question 17 is 3 marks)

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**PAST PAPER SOLUTION****Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Algebraic proof. The tag is acceptable because the question asks for exact-form proof.

**Solution**

$$\begin{aligned}\frac{1 + \sqrt{5}}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}} &= \frac{(1 + \sqrt{5})(3 + \sqrt{5})}{9 - 5} \\ &= \frac{3 + \sqrt{5} + 3\sqrt{5} + 5}{4} \\ &= \frac{8 + 4\sqrt{5}}{4} = 2 + \sqrt{5}\end{aligned}$$

**FINAL ANSWER**

$2 + \sqrt{5}$ , so  $a = 2$  and  $b = 5$ .

**PAST PAPER QUESTION****Q#: 18****Marks: 5**TOPIC: **Differentiation**

May 2024 P1H - May 2024 | P1H

**18** A curve  $C$  has equation  $y = x^3 - 40x + 1$

Find the coordinates of both the points on  $C$  at which the gradient is 8

( ..... , ..... )

( ..... , ..... )

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(Total for Question 18 is 5 marks)

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**PAST PAPER SOLUTION****Q#: 18****Marks: 5****TOPIC: Differentiation**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to differentiation. The question asks for points where the derivative has a given value.

**Solution**

$$y = x^3 - 40x + 1$$

$$\frac{dy}{dx} = 3x^2 - 40$$

The gradient is 8, so

$$3x^2 - 40 = 8$$

$$3x^2 = 48$$

$$x^2 = 16$$

$$x = 4 \quad \text{or} \quad x = -4$$

When  $x = 4$ :

$$y = 4^3 - 40(4) + 1 = -95$$

When  $x = -4$ :

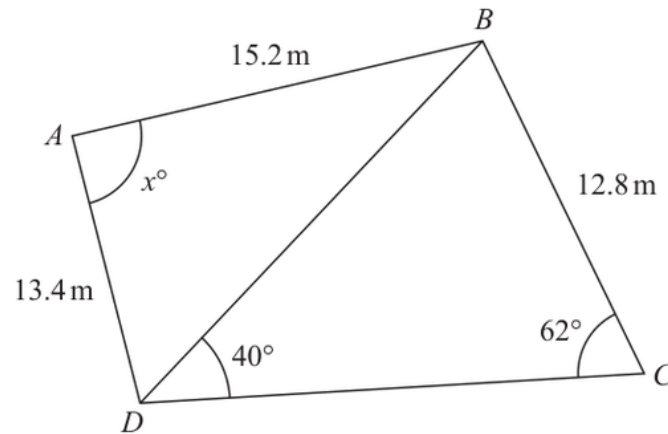
$$y = (-4)^3 - 40(-4) + 1 = 97$$

**FINAL ANSWER**

$(-4, 97)$  and  $(4, -95)$ .

**PAST PAPER QUESTION****Q#: 19****Marks: 5**TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2024 P1H - May 2024 | P1H

**19** Here is quadrilateral  $ABCD$ Diagram **NOT**  
accurately drawn

Work out the value of  $x$   
Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$ **(Total for Question 19 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 19** **Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION**

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**Topic check.** Sine, Cosine Rule & Area of Triangles. The tag is correct.**Solution**In triangle  $BCD$ ,

$$\angle DBC = 180^\circ - 40^\circ - 62^\circ = 78^\circ$$

By the sine rule,

$$\frac{DB}{\sin 62^\circ} = \frac{12.8}{\sin 40^\circ}$$

$$DB = 17.582\dots$$

Now use cosine rule in triangle  $ABD$ :

$$\cos x = \frac{15.2^2 + 13.4^2 - 17.582\dots^2}{2(15.2)(13.4)}$$

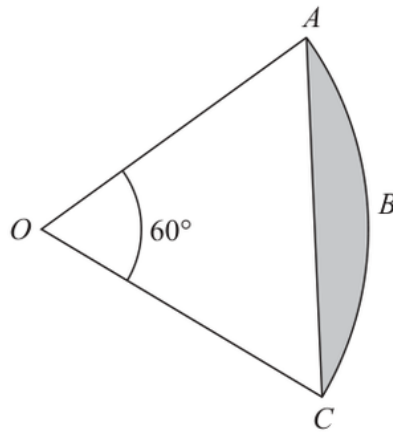
$$x = 75.577\dots^\circ$$

**FINAL ANSWER**

75.6°.

**PAST PAPER QUESTION****Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

**20** The diagram shows a sector  $OABC$  of a circle centre  $O$ Diagram **NOT**  
accurately drawnAngle  $AOC = 60^\circ$ The area of the shaded segment  $ABC$  is  $38 \text{ cm}^2$ Work out the perimeter of the shaded segment  $ABC$ 

Give your answer correct to one decimal place.

..... cm

**(Total for Question 20 is 4 marks)**



**PAST PAPER SOLUTION****Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**Let the radius of the sector be  $r$ .

The shaded segment is

sector area – triangle area

The angle is  $60^\circ$ , so

$$38 = \frac{60}{360}\pi r^2 - \frac{1}{2}r^2 \sin 60^\circ$$

$$38 = r^2 \left( \frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)$$

$$r = 20.4815\dots$$

The perimeter of the shaded segment is the arc  $AC$  plus the chord  $AC$ .

Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

Chord length:

$$2r \sin 30^\circ = r$$

So the perimeter is

$$\begin{aligned} & \frac{\pi r}{3} + r \\ &= \frac{\pi(20.4815\dots)}{3} + 20.4815\dots \\ &= 41.9296\dots \end{aligned}$$

**FINAL ANSWER**

41.9 cm

**PAST PAPER QUESTION****Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

20 The diagram shows a sector  $OABC$  of a circle centre  $O$

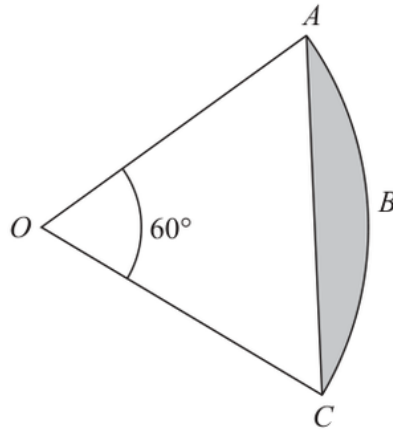


Diagram **NOT**  
accurately drawn

Angle  $AOC = 60^\circ$

The area of the shaded segment  $ABC$  is  $38 \text{ cm}^2$

Work out the perimeter of the shaded segment  $ABC$

Give your answer correct to one decimal place.

..... cm

(Total for Question 20 is 4 marks)

**PAST PAPER SOLUTION****Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**Let the radius of the sector be  $r$ .

The shaded segment is

sector area – triangle area

The angle is  $60^\circ$ , so

$$38 = \frac{60}{360}\pi r^2 - \frac{1}{2}r^2 \sin 60^\circ$$

$$38 = r^2 \left( \frac{\pi}{6} - \frac{\sqrt{3}}{4} \right)$$

$$r = 20.4815\dots$$

The perimeter of the shaded segment is the arc  $AC$  plus the chord  $AC$ .

Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

Chord length:

$$2r \sin 30^\circ = r$$

So the perimeter is

$$\begin{aligned} & \frac{\pi r}{3} + r \\ &= \frac{\pi(20.4815\dots)}{3} + 20.4815\dots \\ &= 41.9296\dots \end{aligned}$$

**FINAL ANSWER**

41.9 cm

**PAST PAPER QUESTION****Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

May 2024 P1H - May 2024 | P1H

**21** A curve has equation  $y = f(x)$ 

There is one minimum point on this curve.

The coordinates of this minimum point are  $(5, -4)$ 

Write down the coordinates of the minimum point on the curve with equation

(i)  $y = f(x + 7)$

( ..... , ..... )  
(1)

(ii)  $y = f(x) - 6$

( ..... , ..... )  
(1)**(Total for Question 21 is 2 marks)**

**PAST PAPER SOLUTION****Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**The minimum point of  $y = f(x)$  is  $(5, -4)$ .

For

$$y = f(x + 7)$$

the graph moves 7 units left:

$$(5, -4) \rightarrow (-2, -4)$$

For

$$y = f(x) - 6$$

the graph moves 6 units down:

$$(5, -4) \rightarrow (5, -10)$$

**FINAL ANSWER** $(-2, -4)$  and  $(5, -10)$

**PAST PAPER QUESTION****Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

May 2024 P1H - May 2024 | P1H

**21** A curve has equation  $y = f(x)$ 

There is one minimum point on this curve.

The coordinates of this minimum point are  $(5, -4)$ 

Write down the coordinates of the minimum point on the curve with equation

(i)  $y = f(x + 7)$

( ..... , ..... )  
(1)

(ii)  $y = f(x) - 6$

( ..... , ..... )  
(1)**(Total for Question 21 is 2 marks)**

**PAST PAPER SOLUTION****Q#: 21****Marks: 2****TOPIC: Transformations of Graphs**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**The minimum point of  $y = f(x)$  is  $(5, -4)$ .

For

$$y = f(x + 7)$$

the graph moves 7 units left:

$$(5, -4) \rightarrow (-2, -4)$$

For

$$y = f(x) - 6$$

the graph moves 6 units down:

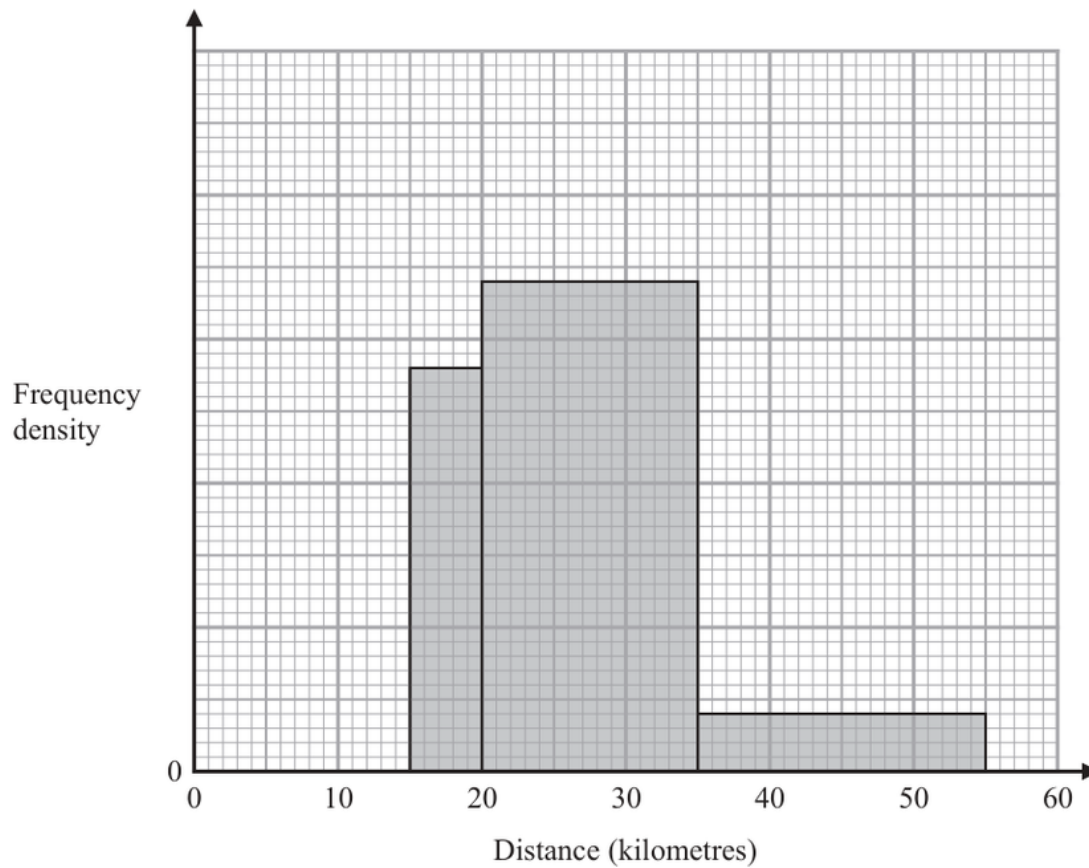
$$(5, -4) \rightarrow (5, -10)$$

**FINAL ANSWER** $(-2, -4)$  and  $(5, -10)$

**PAST PAPER QUESTION****Q#: 22****Marks: 3****TOPIC: Histograms**

May 2024 P1H - May 2024 | P1H

- 22** The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.  
 None of the adults ran more than 55 kilometres.  
 14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(Total for Question 22 is 3 marks)



**PAST PAPER SOLUTION****Q#: 22****Marks: 3**TOPIC: **Histograms**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For  $15 \leq d < 20$ , the frequency is 14 and the class width is 5, so

$$\text{frequency density} = \frac{14}{5} = 2.8$$

Using this scale from the histogram:

$$20 \leq d < 35 \quad \text{has frequency density } 3.4$$

$$35 \leq d \leq 55 \quad \text{has frequency density } 0.4$$

So the known frequencies are

$$14 + (15)(3.4) + (20)(0.4)$$

$$= 14 + 51 + 8 = 73$$

There are 100 adults in total, so the missing frequency from 5 km to 15 km is

$$100 - 73 = 27$$

The missing class width is 10, so its frequency density is

$$\frac{27}{10} = 2.7$$

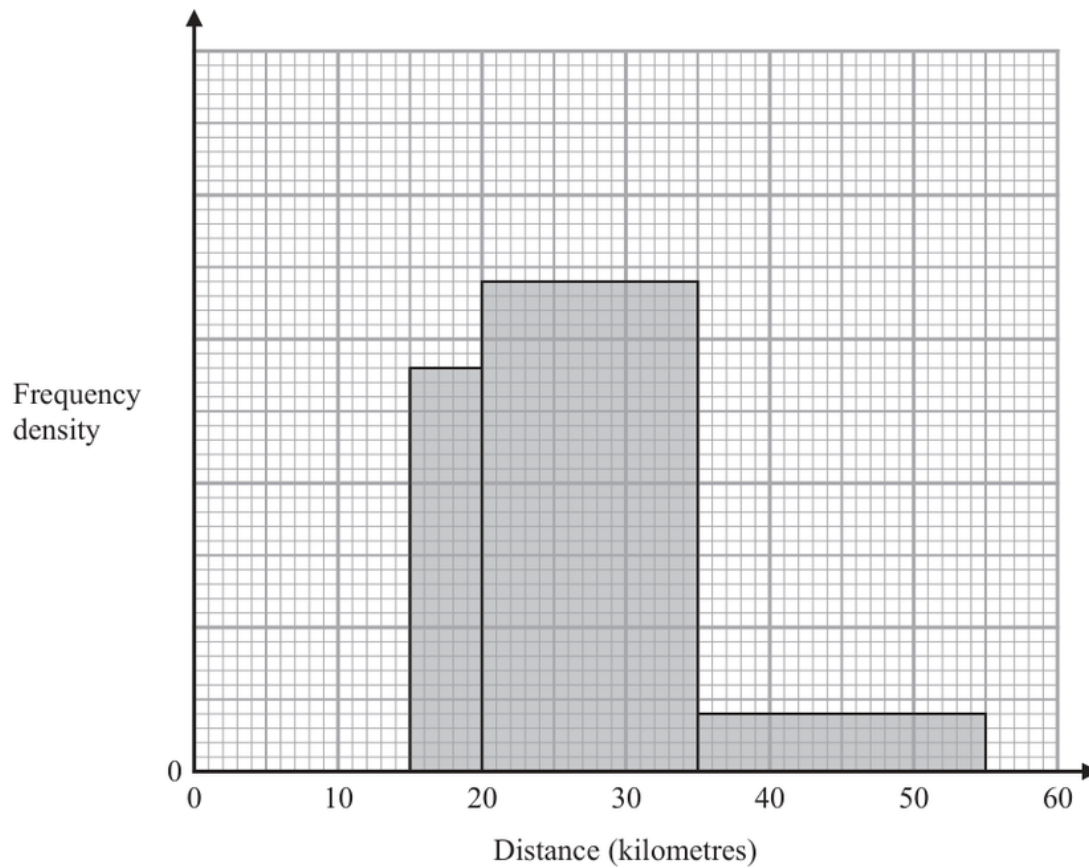
**FINAL ANSWER**

draw the missing bar from 5 to 15 with frequency density 2.7

**PAST PAPER QUESTION****Q#: 22****Marks: 3****TOPIC: Histograms**

May 2024 P1H - May 2024 | P1H

- 22** The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.  
 None of the adults ran more than 55 kilometres.  
 14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(Total for Question 22 is 3 marks)

**PAST PAPER SOLUTION****Q#: 22****Marks: 3**TOPIC: **Histograms**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For  $15 \leq d < 20$ , the frequency is 14 and the class width is 5, so

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Using this scale from the histogram:

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There are 100 adults in total, so the missing frequency from 5 km to 15 km is

$$100 - 73 = 27$$

The missing class width is 10, so its frequency density is

$$\frac{27}{10} = 2.7$$

**FINAL ANSWER**

draw the missing bar from 5 to 15 with frequency density 2.7

**PAST PAPER QUESTION**TOPIC: **Volume & Surface Area**

May 2024 P1H - May 2024 | P1H

**Q#: 23****Marks: 5**

- 23** A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

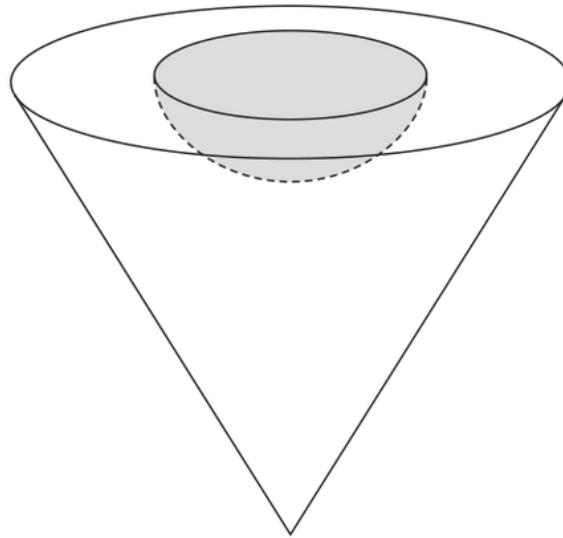


Diagram **NOT**  
accurately drawn

The radius of the hemisphere is  $2x$  cm

The radius of the base of the cone is  $5x$  cm

The vertical height of the cone is  $6x$  cm

The volume of the solid shape is  $6948\pi$  cm<sup>3</sup>

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm<sup>2</sup>

**(Total for Question 23 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 23****Marks: 5****TOPIC: Volume & Surface Area**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**The cone has radius  $5x$  and height  $6x$ .

Volume of the cone:

$$\begin{aligned}\frac{1}{3}\pi(5x)^2(6x) \\ = 50\pi x^3\end{aligned}$$

The removed hemisphere has radius  $2x$ .

Volume of the hemisphere:

$$\begin{aligned}\frac{2}{3}\pi(2x)^3 \\ = \frac{16}{3}\pi x^3\end{aligned}$$

The remaining solid has volume  $6948\pi$ , so

$$\begin{aligned}50\pi x^3 - \frac{16}{3}\pi x^3 &= 6948\pi \\ \frac{134}{3}\pi x^3 &= 6948\pi\end{aligned}$$

Cancel  $\pi$ :

$$\begin{aligned}\frac{134}{3}x^3 &= 6948 \\ x^3 &= \frac{10422}{67}\end{aligned}$$

$$x = 5.3781\dots$$

The solid hemisphere removed has radius  $2x$ .

Total surface area of a solid hemisphere is

$$3\pi r^2$$

So

$$3\pi(2x)^2 = 12\pi x^2$$

$$12\pi(5.3781\dots)^2 = 1090.3903\dots$$

**FINAL ANSWER**

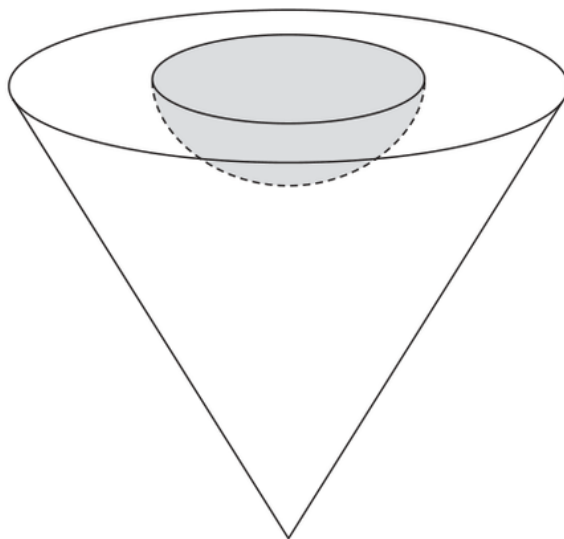
$1090 \text{ cm}^2$

Dr. Eslam Ahmed

**PAST PAPER QUESTION****Q#: 23** **Marks: 5****TOPIC: Volume & Surface Area**

May 2024 P1H - May 2024 | P1H

- 23** A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

Diagram **NOT**  
accurately drawnThe radius of the hemisphere is  $2x$  cmThe radius of the base of the cone is  $5x$  cmThe vertical height of the cone is  $6x$  cmThe volume of the solid shape is  $6948\pi$  cm<sup>3</sup>Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm<sup>2</sup>**(Total for Question 23 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 23****Marks: 5****TOPIC: Volume & Surface Area**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**The cone has radius  $5x$  and height  $6x$ .

Volume of the cone:

$$\begin{aligned}\frac{1}{3}\pi(5x)^2(6x) \\ = 50\pi x^3\end{aligned}$$

The removed hemisphere has radius  $2x$ .

Volume of the hemisphere:

$$\begin{aligned}\frac{2}{3}\pi(2x)^3 \\ = \frac{16}{3}\pi x^3\end{aligned}$$

The remaining solid has volume  $6948\pi$ , so

$$\begin{aligned}50\pi x^3 - \frac{16}{3}\pi x^3 &= 6948\pi \\ \frac{134}{3}\pi x^3 &= 6948\pi\end{aligned}$$

Cancel  $\pi$ :

$$\begin{aligned}\frac{134}{3}x^3 &= 6948 \\ x^3 &= \frac{10422}{67}\end{aligned}$$

$$x = 5.3781\dots$$

The solid hemisphere removed has radius  $2x$ .

Total surface area of a solid hemisphere is

$$3\pi r^2$$

So

$$3\pi(2x)^2 = 12\pi x^2$$

$$12\pi(5.3781\dots)^2 = 1090.3903\dots$$



**FINAL ANSWER**

$1090 \text{ cm}^2$

Dr. Eslam Ahmed

**PAST PAPER QUESTION****Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2024 P1H - May 2024 | P1H

**24** A polygon has  $n$  sides, where  $n > 5$ 

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is  $84^\circ$ The common difference of the sequence is  $4^\circ$ 

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

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**(Total for Question 24 is 6 marks)**

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Dr. Eslam

**PAST PAPER SOLUTION****Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The interior angles form an arithmetic sequence with first term 84 and common difference 4.  
So the sum of the angles is

$$\frac{n}{2}(2(84) + (n - 1)4)$$

$$= \frac{n}{2}(168 + 4n - 4)$$

$$= 2n^2 + 82n$$

The sum of the interior angles of an  $n$ -sided polygon is

$$180(n - 2)$$

So

$$2n^2 + 82n = 180(n - 2)$$

$$2n^2 - 98n + 360 = 0$$

$$n^2 - 49n + 180 = 0$$

$$(n - 45)(n - 4) = 0$$

Since  $n > 5$ ,

$$n = 45$$

Therefore the sum of the interior angles is

$$180(45 - 2) = 7740$$

**FINAL ANSWER**

7740 degrees

**PAST PAPER QUESTION****Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2024 P1H - May 2024 | P1H

**24** A polygon has  $n$  sides, where  $n > 5$ 

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is  $84^\circ$ The common difference of the sequence is  $4^\circ$ 

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

---

**(Total for Question 24 is 6 marks)**

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Dr. Eslam

**PAST PAPER SOLUTION****Q#: 24****Marks: 6**TOPIC: **Sequences**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

The interior angles form an arithmetic sequence with first term 84 and common difference 4.  
So the sum of the angles is

$$\frac{n}{2}(2(84) + (n - 1)4)$$

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$$(n - 45)(n - 4) = 0$$

Since  $n > 5$ ,

$$n = 45$$

Therefore the sum of the interior angles is

$$180(45 - 2) = 7740$$

**FINAL ANSWER**

7740 degrees

**PAST PAPER QUESTION****Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

**25**  $f(x) = 17 - 3x^2 + 12x$

Write  $f(x)$  in the form  $a - b(x - c)^2$  where  $a$ ,  $b$  and  $c$  are constants.

$f(x) = \dots\dots\dots$

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**(Total for Question 25 is 4 marks)**

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Dr. Eslam Ahmed

**PAST PAPER SOLUTION****Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$\begin{aligned}f(x) &= 17 - 3x^2 + 12x \\&= -3x^2 + 12x + 17 \\&= -3(x^2 - 4x) + 17 \\&= -3((x - 2)^2 - 4) + 17 \\&= -3(x - 2)^2 + 12 + 17 \\&= 29 - 3(x - 2)^2\end{aligned}$$

**FINAL ANSWER**

$$f(x) = 29 - 3(x - 2)^2$$

**PAST PAPER QUESTION****Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

**25**  $f(x) = 17 - 3x^2 + 12x$

Write  $f(x)$  in the form  $a - b(x - c)^2$  where  $a$ ,  $b$  and  $c$  are constants.

$f(x) = \dots\dots\dots$

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**(Total for Question 25 is 4 marks)**

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Dr. Eslam Ahmed



**PAST PAPER SOLUTION****Q#: 25****Marks: 4****TOPIC: Completing the Square**

May 2024 P1H - May 2024 | P1H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$\begin{aligned}f(x) &= 17 - 3x^2 + 12x \\&= -3x^2 + 12x + 17 \\&= -3(x^2 - 4x) + 17 \\&= -3((x - 2)^2 - 4) + 17 \\&= -3(x - 2)^2 + 12 + 17 \\&= 29 - 3(x - 2)^2\end{aligned}$$

**FINAL ANSWER**

$$f(x) = 29 - 3(x - 2)^2$$